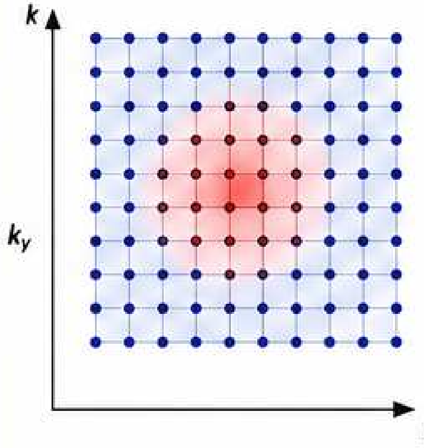
# CENTER vs. OUTER K-SPACE – CONTRAST vs. DETAIL

*Different parts of k-space control different aspects of the image.*

**THE BIG IDEA**
Low spatial frequencies (center) control overall signal and contrast.
High spatial frequencies (outer region) control edges and fine detail.
You need ALL of k-space for the best image.

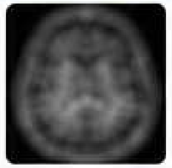
## K-SPACE REGIONS (2D)



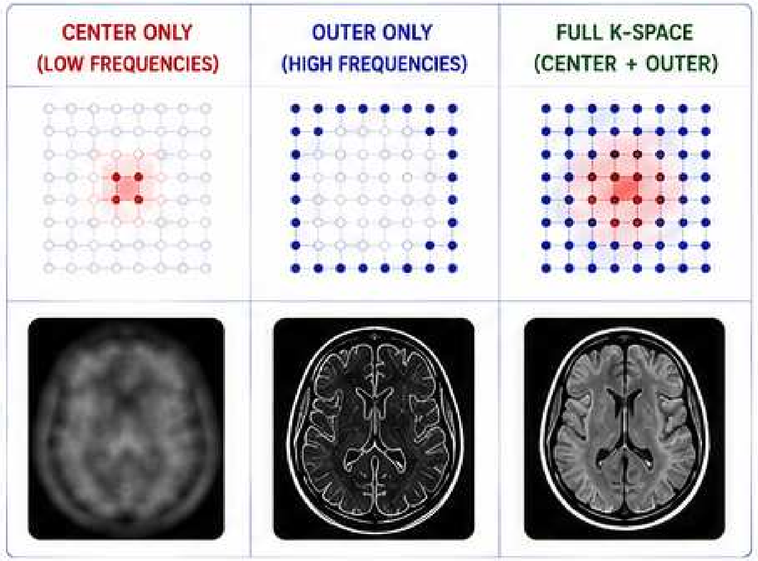

**CENTER (LOW FREQUENCIES)**
- Determines overall signal & contrast
- Changes affect brightness and contrast

**OUTER REGION (HIGH FREQUENCIES)**
- Determines edges, sharpness & fine detail
- Changes affect detail and resolution

## WHAT EACH REGION CONTRIBUTES

| REGION | PRIMARY ROLE | EFFECT ON IMAGE | EXAMPLE |
|---|---|---|---|
| Center Only (Keep center, remove outer) | Overall signal & contrast | Good overall contrast<br>Poor detail<br>Looks blurry |  |
| Outer Only (Remove center, keep outer) | Edges & fine detail | Good edges<br>No contrast<br>Hard to interpret |  |
| Full K-Space (Center + outer) | Both contrast and detail | Good contrast<br>Good detail<br>Best image |  |

## VISUAL DEMONSTRATION

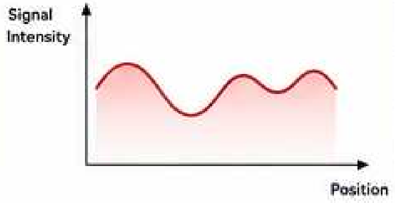



## WHY? THE PHYSICS BEHIND IT

**LOW SPATIAL FREQUENCIES (CENTER)**

Vary slowly across the object → represent broad intensity variations.

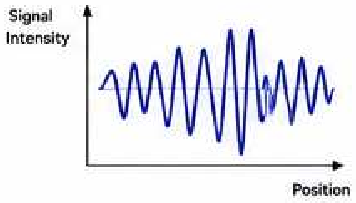
These determine overall signal level and contrast between tissues.




**HIGH SPATIAL FREQUENCIES (OUTER)**

Vary rapidly across the object → represent rapid changes and fine detail.

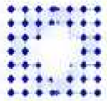
These define edges, sharpness, and small structures.

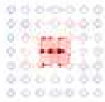
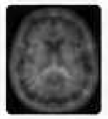


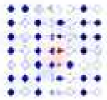

Think of k-space like sound: low frequencies = bass (overall impression), high frequencies = treble (detail). You need the full range for the best experience.

## WHAT HAPPENS IF YOU MISS DATA?

Missing data in k-space causes predictable problems in the image.

 **Miss Center** (DC Offset Error) →  Severe shading and contrast loss

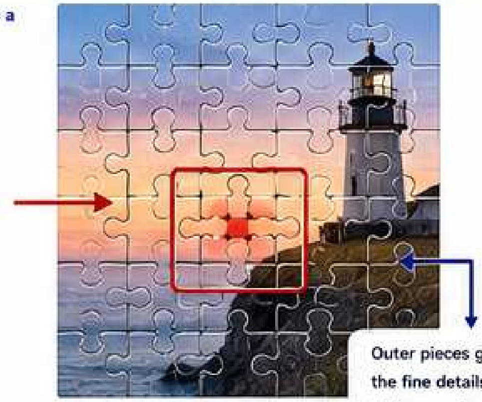
 **Miss Outer** (Low Resolution) →  Blurry image Poor detail

 **Random Missing** (Motion / Dropouts) →  Streaks, dots, noise artifacts

## REAL-WORLD ANALOGY

Think of k-space as a photograph made from puzzle pieces.

Center pieces give you the big picture, colors and overall brightness.



Outer pieces give you the fine details— textures, edges, small structures.

Missing center = wrong colors and contrast.
Missing outer pieces = blurry, no detail.
All pieces together = complete picture.

## KEY TAKEAWAYS

- Center k-space controls contrast and overall signal.
- Outer k-space controls edges and fine detail.
- Losing center → poor contrast, shading.
- Losing outer → blurry, no detail.
- Full k-space = best image quality.

**BOTTOM LINE** Center = Contrast · Outer = Detail · Full K-Space = Best Image

**WHAT'S NEXT?** Now that you know what different parts of k-space do, let's see how k-space relates to resolution, FOV, matrix size, and bandwidth in the next poster.